# SANSKRUTI NAIK

**AI / ML Engineer — Generative AI · RAG · Computer Vision**

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Portfolio · GitHub · LinkedIn

## SUMMARY

AI/ML engineer (B.Tech AI & ML, graduating October 2026) who takes AI from notebook to production: private, citation-grounded RAG systems, explainable computer vision and secure real-time applications. Owns the full path from data and model to API, mobile client and stakeholder walkthrough. Seeking an AI/ML or GenAI engineering role on products people rely on.

## EXPERIENCE

**AI / ML Intern** — Endureva Innovations LLP — Jan 2026 – Jun 2026

- Shipped an AI crop-disease diagnosis feature end to end: data pipeline, CNN training, a **FastAPI** service and a **Flutter** app built for field use (**38 diseases, 92% accuracy**).
- Made it fast enough for real-world use: **under 200 ms** per prediction, and **30% faster** inference after quantization and batching.
- Built trust into the product with **Grad-CAM** heatmaps that show users which part of the leaf drove each diagnosis.
- Built an offline-first restaurant **POS** (Flutter, FastAPI, SQLite) with role-based access, inventory alerts and live sales analytics (code).
- Turned model results and trade-offs into clear recommendations for non-technical stakeholders.

## PROJECTS

**Tender Intelligence — Offline RAG Platform** · Live app — 2026 – present
Python · LangChain · FastAPI · Ollama · ChromaDB · SQLite FTS5 · MCP

- Designed the retrieval pipeline and LLM prompting that answer eligibility, EMD, scope, risk and penalty questions with **clause-level citations (source sentence + page)**, cutting tender analysis **from hours to minutes**.
- Combined vector and keyword search with rank fusion and **cross-encoder reranking**; measured recall, rank and latency against a golden question set.
- Built Go/No-Go, risk-finder and contradiction-finder modules on the retrieval layer, plus a Redmine integration via **MCP**; runs fully offline for confidential documents.

**SecureChat — End-to-End Encrypted Messaging** — 2026 – present
React 18 · XMPP/ejabberd · OMEMO2 · WebRTC · PostgreSQL · Docker · Capacitor

- Architected **OMEMO2 (Signal Protocol)** encryption across messages, media and calls, with a local-first IndexedDB store for offline history.
- Implemented WebRTC voice/video via TURN/STUN and encrypted file sharing **up to 500 MB** using chunked AES-GCM streaming; shipped Android and iOS builds from one codebase.

**Unauthorised Construction Detection (Geospatial AI)** — 2026 – present
Python · YOLOv8 · Sentinel-1/2 · SAR · PostGIS

- Architected the PostGIS spatial database and Web GIS dashboard for a three-model ensemble fusing Sentinel-2, SAR and Cartosat-3 imagery, with a five-day automated alert cycle. Targets: ~95% accuracy, <4% false positives (in progress).

**XAI-Care — Explainable Multimodal Fall Detection** — 2025 – 2026
TensorFlow · FastAPI · Grad-CAM · SHAP · Flutter

- Built the multimodal fusion logic (wearable sensors + camera) and a caregiver dashboard where Grad-CAM and SHAP explain every alert.

**Also built:** Water Survey, offline-first Flutter field app · 3D portfolio with a motion-captured avatar, Three.js (live, code) · Sign-language learning app (Smart India Hackathon)

## TECHNICAL SKILLS

**Generative AI:** LLMs, RAG, LangChain, OpenAI API, Hugging Face, Ollama, prompt engineering, vector embeddings, ChromaDB, hybrid search, reranking, MCP

**Machine Learning & Vision:** TensorFlow, PyTorch, scikit-learn, CNNs, YOLOv8, OpenCV, model quantization, Grad-CAM, SHAP, NLP

**Languages:** Python, JavaScript, SQL, Dart, R, HTML/CSS · **Backend & Data:** FastAPI, REST APIs, PostgreSQL, PostGIS, SQLite, JWT

**Data & BI:** Power BI, SQL, data visualisation · **Apps & Real-time:** React, Flutter, Android, Capacitor, WebRTC, XMPP, Signal Protocol

**DevOps & Tools:** Docker, Docker Compose, Nginx, Git, Vercel

## EDUCATION

**B.Tech, Artificial Intelligence & Machine Learning** — Sanjay Ghodawat University · CGPA 8.36 / 10 — Oct 2026

**12th Standard, Science** — Vivekanand College, Kolhapur · **Languages:** English, Hindi, Marathi (fluent), German (basic) — 2022

## CERTIFICATIONS, LEADERSHIP & AWARDS

- **Certifications:** NPTEL AI for Management (12 weeks, 2026, 82/100) · Cloud & Edge Computing workshop, SGU with Prof. Rajkumar Buyya (Univ. of Melbourne), 2022 · ISRO Remote Sensing & Digital Image Analysis · Accenture Software Engineering Simulation · C++ OOP
- **3rd Prize, Smart India Hackathon** (sign-language learning app) · **Management Team Lead, Prabal Hackathon** (40 hours, 600+ participants) · **Core Member, GDG SGU**